

Name: _____

8.1 Similar polygons.

1. $\triangle ABC \sim \triangle DEF$, $AB = 32$, $DE = 20$. Give both scale factors.
2. Quadrilaterals with congruent corresponding angles have sides 6, 9, 12, 15 and 8, 12, 16, 20. Similar? Check.
3. Rectangles 16×12 and 20×14 : similar? Explain.
4. $\triangle JKL \sim \triangle PQR$, $JK = 6x$, $PQ = 24$, $KL = 30$, $QR = 20$. Find x .
5. Two similar pentagons have scale factor $\frac{5}{6}$ and the larger has perimeter 72. Find the smaller perimeter.

8.2-8.3 Triangle similarity.

6. Two triangles each contain angles of 44° and 59° . Similar? Third angle?
7. Sides 9, 15, 21 and 12, 20, 28: similar? Show ratios.
8. Sides 8 and 14 with a 33° included angle; sides 12 and 21 with a 33° included angle. Similar? Why?
9. In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$, $AD = 14$, $DB = 7$, $AE = 18$. Find EC .
10. A 5-ft mailbox post casts a 4-ft shadow while a building casts a 52-ft shadow. Find the building's height.

8.4 Proportionality theorems.

11. $\overline{XY} \parallel \overline{AC}$ with $BX = 18$, $XA = 12$, $BY = 21$. Find YC .
12. Pieces 16, 20 on one side and 20, 25 on the other: is the segment parallel to the third side?
13. Three parallel lines cut transversals into 24, 32 (left) and x , 28 (right). Find x .
14. In $\triangle ABC$, $\overline{BD}$ bisects $\angle B$, $AB = 33$, $BC = 22$, $AC = 30$. Find AD and DC .
15. In $\triangle JKL$, $\overline{KM}$ bisects $\angle K$, $JM = x$, $ML = x - 3$, $JK = 28$, $KL = 21$. Find x and JL .

Putting it together.

16. $\triangle RST \sim \triangle UVW$, $RS = 18$, $ST = 24$, $TR = 30$, $UV = 12$. Find the perimeter of $\triangle UVW$.
17. A drawing of a 45-ft sailboat uses scale factor $1/60$. How long is the drawing in inches?
18. Why is there no 'AAA congruence' but there IS an 'AA similarity'? One sentence each.
19. In the SSA trap for similarity (two proportional sides, congruent NON-included angle), what goes wrong?
20. State the two 'free ratio' facts that come with every pair of similar polygons.